# Trading rule discovery in the US stock market: An empirical study

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# Abstract(1/2)

This study develops a new template grid - rounding top and saucer- to detect buy signals. Most of the previous studies utilize historical data to derive the template grid, but do not clearly explain how to format weight values of the template grid. This makes the template a ''black box'' for users since it is difficult to infer the process of the template information. Therefore, this study proposes a simple and explicit method for deriving the template grid. In addition, to more accurately detect buy signals, the trading rules are developed by capturing reversal of price trend.

# Abstract(2/2)

The empirical results indicate that the template grid and the proposed trading rules developed in this study have considerable forecasting power across tech stocks in the US, including MSFT, IBM, INTC, ORCL, DELL, APPLE and HP, since the average returns of the proposed trading rule are greater than that the results of buying every day over the sample period. The method proposed here could therefore become an effective component of an expert system of assist investor in investment decisions.

# Template Grid

| UP | | | Horizon | | | | Down | | |
|---|---|---|---|---|---|---|---|---|---|
| -1.000 | -0.538 | 0.375 | 1.000 | 1.000 | 1.000 | 1.000 | 0.375 | -0.538 | -1.000 |
| -0.778 | 0.231 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.231 | -0.778 |
| -0.556 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | -0.556 |
| -0.333 | 1.000 | 1.000 | 1.000 | 0.643 | 0.643 | 1.000 | 1.000 | 1.000 | -0.333 |
| -0.111 | 1.000 | 1.000 | 0.524 | 0.286 | 0.286 | 0.524 | 1.000 | 1.000 | -0.111 |
| 0.111 | 1.000 | 0.375 | 0.048 | -0.071 | -0.071 | 0.048 | 0.375 | 1.000 | 0.111 |
| 0.333 | 0.231 | -0.250 | -0.429 | -0.429 | -0.429 | -0.429 | -0.250 | 0.231 | 0.333 |
| 0.556 | -0.538 | -0.875 | -0.905 | -0.786 | -0.786 | -0.905 | -0.875 | -0.538 | 0.556 |
| 0.778 | -1.308 | -1.500 | -1.381 | -1.143 | -1.143 | -1.381 | -1.500 | -1.308 | 0.778 |
| 1.000 | -2.077 | -2.125 | -1.857 | -1.500 | -1.500 | -1.857 | -2.125 | -2.077 | 1.000 |

**Fig. 1.** Weights representing rounding top pattern variation ($T_1$).

Rounding top pattern

| Down | | | Horizon | | | | UP | | |
|---|---|---|---|---|---|---|---|---|---|
| 1.000 | -2.077 | -2.125 | -1.857 | -1.500 | -1.500 | -1.857 | -2.125 | -2.077 | 1.000 |
| 0.778 | -1.308 | -1.500 | -1.381 | -1.143 | -1.143 | -1.381 | -1.500 | -1.308 | 0.778 |
| 0.556 | -0.538 | -0.875 | -0.905 | -0.786 | -0.786 | -0.905 | -0.875 | -0.538 | 0.556 |
| 0.333 | 0.231 | -0.250 | -0.429 | -0.429 | -0.429 | -0.429 | -0.250 | 0.231 | 0.333 |
| 0.111 | 1.000 | 0.375 | 0.048 | -0.071 | -0.071 | 0.048 | 0.375 | 1.000 | 0.111 |
| -0.111 | 1.000 | 1.000 | 0.524 | 0.286 | 0.286 | 0.524 | 1.000 | 1.000 | -0.111 |
| -0.333 | 1.000 | 1.000 | 1.000 | 0.643 | 0.643 | 1.000 | 1.000 | 1.000 | -0.333 |
| -0.556 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | -0.556 |
| -0.778 | 0.231 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.231 | -0.778 |
| -1.000 | -0.538 | 0.375 | 1.000 | 1.000 | 1.000 | 1.000 | 0.375 | -0.538 | -1.000 |

**Fig. 2.** Weights representing saucer pattern variation ($T_2$).

Saucer pattern

# How to get the weight values of the template grid?

step1 ： the pattern mapped to the corresponding cells $w_{ij}$ denotes as 1

step2 ： the weight values for each column are calculated at a linear decrement and the sum of weight values for all cells in each column should be equal to zero.

| a | | | | | | | | | | b | c |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | 1 | 1-5d | -2.125 |
| | | | | | | | | | | 1-4d | -1.500 |
| | | | | | | | | | | 1-3d | -0.875 |
| | | | | | | | | | | 1-2d | -0.250 |
| | 1 | | | | | | | 1 | | 1-d | 0.375 |
| | 1 | 1 | | | | | 1 | 1 | | 1 | 1.000 |
| | 1 | 1 | 1 | | | 1 | 1 | 1 | | 1 | 1.000 |
| | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | 1 | 1.000 |
| | | 1 | 1 | 1 | 1 | 1 | 1 | | | 1 | 1.000 |
| | | | 1 | 1 | 1 | 1 | | | | 1-d | 0.375 |

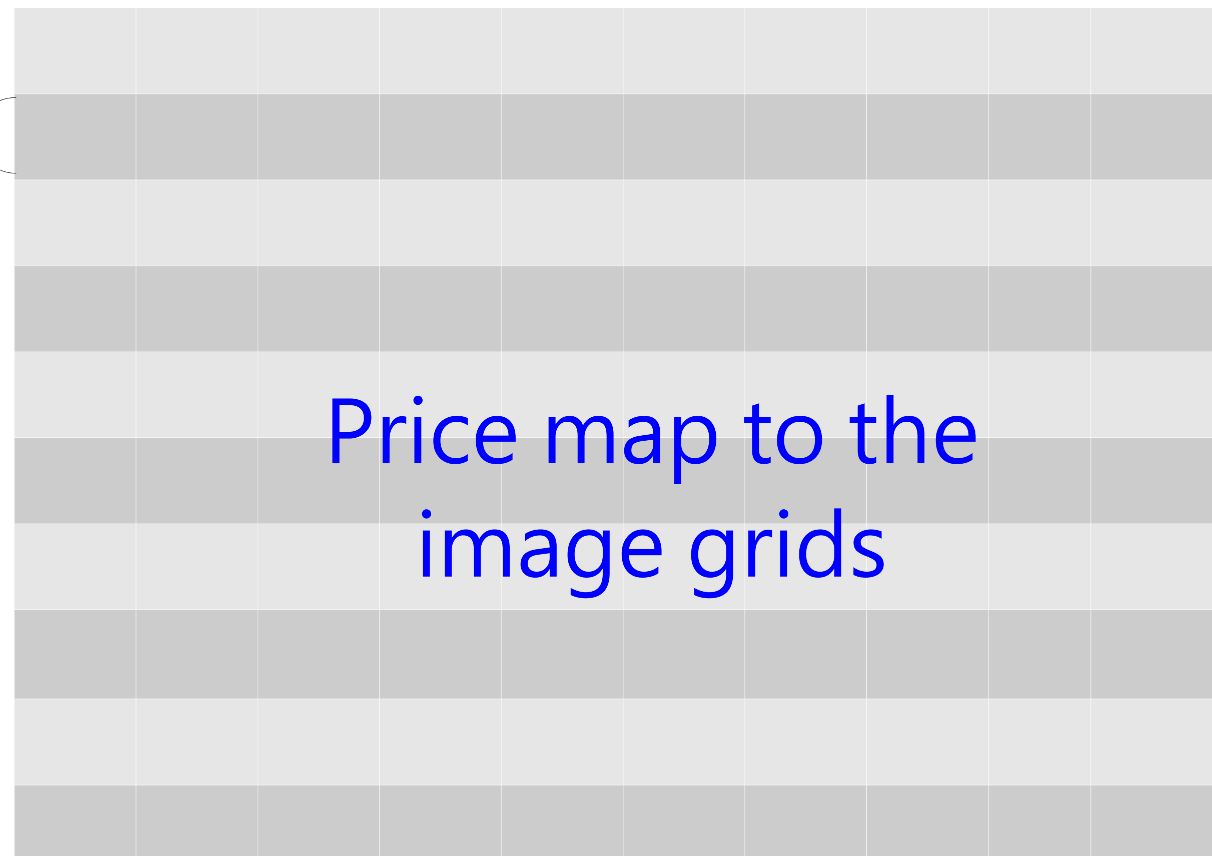
$$\text{Sum of the third column} = (1-5d)+(1-4d)+(1-3d)+(1-2d)+(1-d)+1+1+1+1+(1-d)=0$$

$$d = 0.625$$

# Mapping w-days price into image 10 x10 image grids

In trading day t, we take w-days of daily price values and map into a 10x10 images grids for the fitting window ending with trading day (t-1).

$$\frac{(P_w^{max} - P_w^{min})}{10}$$




w/10

$$g_{ij} = \frac{f_{ij}}{w/10}$$

Where $f_{ij}$ is the frequency with which the w/10 daily price values for column j fall within interval i

| day | price | i |
|---|---|---|
| 1 | 7 | |
| 2 | 11 | |
| 3 | 15 | |
| 4 | 19 | |
| 5 | 22 | |
| 6 | 28 | |
| 7 | 24 | |
| 8 | 35 | |
| 9 | 39 | |
| 10 | 37 | |
| 11 | 41 | |
| 12 | 42 | |
| 13 | 40 | |
| 14 | 46 | |
| 15 | 47 | |
| 16 | 46 | |
| 16 | 46 | |
| 17 | 43 | |
| 18 | 42 | |
| 19 | 39 | |
| 20 | 35 | |

In trading day 21, w= 20

Max price = 47, min price = 7

Interval = (47-7)/10 = 4

| Day i | 1-2 | 3-4 | 5-6 | 7-8 | 9-10 | 11-12 | 13-14 | 15-16 | 17-18 | 19-20 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | 0.5 | 1 | 0.5 | |
| 2 | | | | | 0.5 | 1 | 0.5 | | 1 | 0.5 |
| 3 | | | | 0.5 | 1 | | | | | 1 |
| 4 | | | | 0.5 | | | | | | 0.5 |
| 5 | | | 0.5 | | | | | | | |
| 6 | | | | 0.5 | | | | | | |
| 7 | | 0.5 | 0.5 | | | | | | | |
| 8 | | 1 | | | | | | | | |
| 9 | 0.5 | 0.5 | | | | | | | | |
| 10 | 1 | | | | | | | | | |

# Calculate the fit value

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| -1.000 | -0.538 | 0.375 | 1.000 | 1.000 | 1.000 | 1.000 | 0.375 | -0.538 | -1.000 |
| -0.778 | 0.231 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.231 | -0.778 |
| -0.556 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | -0.556 |
| -0.333 | 1.000 | 1.000 | 1.000 | 0.643 | 0.643 | 1.000 | 1.000 | 1.000 | -0.333 |
| -0.111 | 1.000 | 1.000 | 0.524 | 0.286 | 0.286 | 0.524 | 1.000 | 1.000 | -0.111 |
| 0.111 | 1.000 | 0.375 | 0.048 | -0.071 | -0.071 | 0.048 | 0.375 | 1.000 | 0.111 |
| 0.333 | 0.231 | -0.250 | -0.429 | -0.429 | -0.429 | -0.429 | -0.250 | 0.231 | 0.333 |
| 0.556 | -0.538 | -0.875 | -0.905 | -0.786 | -0.786 | -0.905 | -0.875 | -0.538 | 0.556 |
| 0.778 | -1.308 | -1.500 | -1.381 | -1.143 | -1.143 | -1.381 | -1.500 | -1.308 | 0.778 |
| 1.000 | -2.077 | -2.125 | -1.857 | -1.500 | -1.500 | -1.857 | -2.125 | -2.077 | 1.000 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0.5 | 1 | 0.5 | 0 | 1 | 0.5 |
| 0 | 0 | 0 | 0.5 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0.5 | 0 | 0 | 0 | 0 | 0 | 0.5 |
| 0 | 0 | 0.5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0.5 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0.5 | 0.5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0.5 | 0.5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | | | | | | | | |

**Fig. 1.** Weights representing rounding top pattern variation ($T_1$).

$$Fit_{w,t} = \sum_{i=1}^{10} \sum_{j=1}^{10} (w_{ij} g_{ij})$$

(T1) ： $Fit_{20,20}$ = (1*1) + (0.5*0.778+0.5*-1.308) + (1*-0.538)+(0.5*0.231+0.5*-0.250) +
(0.5*0.048)+(0.5*1)+(0.5*0.643+0.5*-0.333)+(0.5*1+1*1+1*-0.556)+
(0.5*1+1*1+0.5*1+1*0.231+0.5*-0.778)+(0.5*1+1*0.375+0.5*-0.538)
= 4.2585

Higher value indicates better price fit for a rounding top/saucer template!!

# Trading Rule A

If($Depth_{w,t}$ > Threashold$_{Depth}$ x avg($Depth_w$) )and
(Fit$_{w,t}$ > Threashold$_{Fit}$)
then buy and hold for q days

$$Depth_{w,t} = \frac{P^{max}_{w,t} - P^{min}_{w,t}}{P^{min}_{w,t}} \qquad avg\left(Depth_w\right) = \frac{\sum_{t=1}^{T/2} Depth_{w,t}}{T/2}, T \text{ is the full sample period for selected stocks.}$$

If the depth of the pattern is too small, the charting pattern may hold no meaningful information for investment decisions.

# Trading Rule B

If pt > $MA_{w,t}$
    Trading Rule A for template T1(rounding top)
else
    Trading Rule A for template T2(saucer)

$$MA_{k,t} = \frac{\sum_{x=t-k}^{t-1} P_x}{k}$$

To reduce the incidence of the false signals for Trading Rule A and to more accurately capture the price reversal, a different time span(k) of the MA is applied.

# Training the thresholds

**Table 1**
Thresholds with stable profits

| Fitting window size (w) | $Threshold_{Fit}$ | $Threshold_{Depth}$ | Frequency of stable profit |
|---|---|---|---|
| *Panel A: thresholds for rounding top $T_1$* | | | |
| 60 | 5 | 1 | 7 |
| 60 | 6 | 1 | 7 |
| 240 | 5.5 | 1 | 7 |
| 240 | 6 | 1 | 8 |
| 240 | 7 | 0.75 | 7 |
| *Panel B: thresholds for saucer $T_2$* | | | |
| 40 | 5 | 1 | 6 |
| 40 | 5.5 | 1 | 6 |
| 100 | 5 | 1 | 7 |
| 120 | 5 | 1 | 7 |
| 120 | 5.5 | 1 | 7 |
| 120 | 7.5 | 0.25 | 9 |
| 120 | 7.5 | 0.5 | 9 |
| 160 | 5.5 | 1 | 7 |
| 160 | 6 | 1 | 7 |
| 160 | 6.5 | 1 | 7 |
| 200 | 6 | 1 | 7 |
| 200 | 6.5 | 1 | 7 |
| 240 | 6 | 0.75 | 7 |
| 240 | 6 | 1 | 8 |
| 240 | 6.5 | 0.75 | 8 |
| 240 | 6.5 | 1 | 8 |
| 240 | 7.5 | 0.25 | 8 |
| 240 | 7.5 | 0.5 | 8 |

"Frequency of stable profit" means the numbers of holding periods out of 9 ($q$ = 20, 40, 60, 80, 100, 120, 160, 200, 240 days) with average annualized returns, after considering the trading cost (1%), of over 20%.

- Employs NASDAQ pre period(divide 02/05/1971 to 09/24/2007 into two equal sub-periods).
- Use Trading Rule A to find various sets of thresholds
- For 9 holding periods(q=20, 40, 60, 80, 100, 120, 160, 200, 240 days), more than 5 holding periods should have an average annualized return after considering the trading cost(1%) of over 20%.

# Experimental results

- 23 filter rules are employed in the Trading rule B to detect buy signals
- If the phenomena satisfy one of the filter rules, investors should buy on that trading day and hold for q days, and then sell.
- Use MSFT, IBM, INTC, ORCL, DELL, APPLE, HP to avoid suspicions of data mining.

$$\text{Market Average Return} = \frac{\sum_{t=m}^{n} \left[ (P_{t+q} - P_t)/P_t \right]}{n - m + 1}$$

$$\text{Number of Buys} = \sum_{t=m}^{n} B_t$$

$B_t = 1$, if the phenomena satisfy Trading Rule B

$B_t = 0$, otherwise

$$\text{Trading Rule Average Return} = \frac{\sum_{t=m}^{n} \left[ (P_{t+q} - P_t) \cdot B_t / P_t \right]}{\sum_{t=m}^{n} B_t}$$

**Table 2**
Results of Trading Rule B for buy and hold for 180 days during the pre and post period across tech stocks

| *k* | MSFT | | IBM | | INTC | | ORCL | | DELL | | APPLE | | HP | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | pre | post | pre | post | pre | post | pre | post | pre | post | pre | post | pre | post |
| 20 | 7.48 (598) | 4.02 (253) | 15.93 (533) | 4.17 (1234) | 26.42 (360) | 26.76 (712) | 78.89 (436) | 23.70 (690) | 69.63 (544) | −0.20 (453) | 14.79 (541) | 37.07 (539) | 22.29 (656) | 3.51 (579) |
| 40 | 8.27 (436) | −7.50 (178) | 11.83 (348) | 8.85 (945) | 37.59 (304) | 33.98 (611) | 90.52 (393) | 21.77 (530) | 66.00 (381) | 5.18 (370) | 19.35 (499) | 51.45 (358) | 32.05 (439) | −3.50 (426) |
| 60 | 12.43 (387) | −5.70 (172) | 9.55 (293) | 6.89 (959) | 39.62 (279) | 33.91 (592) | 93.90 (380) | 17.96 (547) | 54.88 (336) | 5.02 (311) | 21.49 (440) | 48.09 (235) | 30.52 (407) | −9.90 (416) |
| 120 | 10.62 (289) | −5.80 (195) | 9.59 (314) | 4.50 (991) | 34.06 (325) | 35.62 (653) | 83.98 (371) | 15.46 (676) | 52.21 (384) | 4.07 (287) | 20.26 (378) | 43.49 (277) | 27.13 (482) | 3.51 (526) |
| 180 | 7.24 (523) | −7.30 (246) | 17.12 (423) | 5.05 (1074) | 20.47 (446) | 33.96 (924) | 68.44 (450) | 8.48 (807) | 52.42 (509) | 3.02 (481) | 15.73 (533) | 32.25 (301) | 33.99 (663) | 11.10 (625) |
| 240 | 6.08 (691) | −8.50 (293) | 16.82 (626) | 5.95 (1299) | 19.19 (485) | 34.48 (1018) | 78.07 (704) | 4.36 (854) | 64.08 (667) | 1.64 (645) | 17.68 (682) | 29.00 (374) | 31.60 (928) | 12.47 (778) |
| Market | 7.71 | −8.30 | 0.10 | 1.15 | 20.73 | −5.10 | 14.37 | 11.23 | 47.90 | −8.50 | 8.08 | 32.63 | 10.60 | 7.34 |

The sample period of each stock is divided equally into pre and post period. "Market" refers to average market return and is the average profit obtained by buying every day. *Trading Rule B* average return is the average profit obtained by buying only on the rule-indicated days. *k* is the time span for moving average in *Trading Rule B*. The parentheses are the number of buys. Annualized return is calculated on a 240 trading-day-per-year basis.

**Table 3**
Buying run statistics for *Trading Rule B* with 40-day MA

| Stock name | Market | Buying on the days specified by the rule | | | | First day of run only | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | *N* (buys) | Annualized return (%) | Excess profit | | *N* (runs) | Average run length | Annualized return (%) | Excess profit | |
| MSFT | −0.27 | 614 | 3.69 | $3.96^{***}$ | (0.0007) | 106 | 5.79 | 5.62 | $5.89^{*}$ | (0.0750) |
| IBM | 0.62 | 1293 | 9.65 | $9.03^{***}$ | (0.0000) | 246 | 5.26 | 8.04 | $7.42^{***}$ | (0.0000) |
| INTC | 7.80 | 915 | 35.18 | $27.38^{***}$ | (0.0000) | 168 | 5.45 | 21.20 | $13.40^{***}$ | (0.0008) |
| ORCL | 12.80 | 923 | 51.05 | $38.25^{***}$ | (0.0000) | 170 | 5.43 | 39.94 | $27.14^{***}$ | (0.0000) |
| DELL | 19.70 | 751 | 36.04 | $16.34^{***}$ | (0.0000) | 178 | 4.22 | 29.50 | $9.80^{***}$ | (0.0040) |
| APPLE | 20.36 | 857 | 32.76 | $12.40^{***}$ | (0.0000) | 137 | 6.26 | 21.45 | 1.09 | (0.3991) |
| HP | 8.97 | 865 | 14.54 | $5.57^{***}$ | (0.0001) | 197 | 4.39 | 15.31 | $6.34^{**}$ | (0.0435) |

# Thank You!!